

Vaasenk

Web App Product Requirements Document

Version 0.1 | MVP + AI Syllabus Intelligence | Prepared for frontend/backend planning

1. Product Summary

Vaasenk is a classroom productivity platform for schools, colleges, and coaching centers. It reduces time wasted in copying board notes by allowing teachers to upload notes, photos, PDFs, and learning material into structured classrooms. The web app is the control center for admins and teachers: institution setup, syllabus management, classroom creation, AI question paper generation, teacher-only classroom chatbot, notes management, analytics, and subscriptions.

Decision	Locked specification
Product name	Vaasenk
Primary web users	Institution Admin, Teacher, Student (optional web access)
Frontend	Next.js + TypeScript + Tailwind CSS + ShadCN/Radix UI
Backend	Node.js API using Express.js or NestJS
Database	PostgreSQL as system database; pgvector recommended for embeddings when using Postgres-based RAG
AI workflow option	Flowise or Langflow may be used as an AI orchestration service behind the Node.js backend. Client apps must never call Flowise/Langflow directly.
Syllabus ownership	Admin uploads and maps syllabus PDFs during institution/classroom setup. Teachers should not repeatedly upload the same syllabus.
Teacher AI assistant	Each classroom gets a teacher-only AI chatbot trained/retrieved from the mapped syllabus and sample question papers. Student access is future phase.
Question paper generator	Teacher-only feature that uses mapped syllabus + selected portions + sample question papers + marks blueprint.

2. Product Goals

- Reduce classroom time spent copying board notes.
- Give teachers a fast way to upload, organize, and reuse classroom materials.
- Centralize syllabus and sample question paper management at the admin level.
- Help teachers generate quality question papers without missing important portions.
- Provide every classroom with a syllabus-aware AI assistant for teacher planning and doubt preparation.
- Create an institution-friendly SaaS product with recurring subscription revenue.

Non-goals for MVP

- Do not build a full ERP system in MVP.
- Do not include parent login in MVP.
- Do not allow students to chat with the AI assistant in MVP.
- Do not attempt automated grading in MVP.
- Do not build live video classes in MVP.
- Do not make Vaasenk behave like a general WhatsApp/Discord chat room.

3. Target Users and Personas

Persona	Main need	Success definition
Institution Admin	Set up school, users, syllabus, classrooms, and billing.	Can onboard the institution without technical help and monitor usage.
Teacher	Upload notes, access class syllabus, generate question papers, and chat with classroom AI.	Can save preparation time and avoid repeating manual work.
Student	Access notes, download/bookmark material, and receive classroom updates.	Can study from organized notes without copying everything from the board.
Super Admin / Vaasenk Team	Manage institutions, plans, platform health, and support.	Can operate Vaasenk as a scalable SaaS product.

4. Web App Scope

Module	MVP status	Description
Authentication and roles	MVP	Login, registration/invite flow, role-based routing, secure sessions.
Institution setup	MVP	Admin creates institution profile, classes, sections, subjects, and academic year.
Syllabus repository	MVP	Admin uploads common syllabus PDFs such as Samacheer Kalvi, CBSE, or institution-specific syllabus.
Sample paper repository	MVP	Admin uploads previous/sample papers to guide AI question paper pattern and important question coverage.
Classroom management	MVP	Create classrooms, assign teacher, map syllabus, invite students.
Notes management	MVP	Teacher uploads image/PDF/text notes; students view, download, and bookmark.
AI question paper generator	MVP	Teacher generates question paper using mapped syllabus and sample-paper guidance.
Teacher AI classroom chatbot	MVP	Teacher-only RAG chatbot for classroom syllabus/sample papers.
Notifications	MVP	New notes, new papers, classroom invites, AI processing completion.
Analytics	Phase 2	Student engagement, teacher uploads, subject-wise usage.
Payments/subscriptions	Phase 2/MVP-light	Plan management can start as manual billing; payment gateway later.

5. Role and Permission Matrix

Action	Super Admin	Institution Admin	Teacher	Student
Create institution	Yes	No	No	No
Manage subscription	Yes	Yes	No	No
Create academic year/classes/sections	No	Yes	No	No
Upload syllabus PDFs	No	Yes	No by default	No
Upload sample question papers	No	Yes	Optional with admin permission	No
Create classroom	No	Yes	Yes if allowed	No
Map syllabus to classroom	No	Yes	Request-only or allowed	No
Upload notes to classroom	No	No	Yes	No by default
View notes	No	Yes	Yes	Yes if enrolled
Generate question paper	No	No	Yes	No
Use classroom AI chatbot	No	No	Yes	No in MVP
Ask doubts under notes	No	No	Reply only	Yes

Delete classroom content	No	Yes	Own content only	No
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6. Recommended System Architecture

- Next.js web app communicates only with the Node.js backend API.
- Node.js backend owns authentication, authorization, institution isolation, file metadata, classroom permissions, and billing status.
- PostgreSQL stores users, roles, classrooms, enrollments, syllabus metadata, sample paper metadata, notes, generated papers, chatbot sessions, notifications, and audit logs.
- Object storage stores large files: syllabus PDFs, sample papers, note images/PDFs, generated question papers, exports.
- Redis + BullMQ is recommended for long-running jobs such as PDF parsing, OCR, chunking, embedding, AI paper generation, and notification dispatch.
- AI orchestration can be handled by Flowise/Langflow or a custom Node/Python RAG microservice. The backend must call the AI service server-side.
- For RAG storage, use PostgreSQL + pgvector if the team wants fewer infrastructure pieces. Alternatively, use Qdrant/Pinecone later if scale demands it.
- Every AI request must include institution_id, classroom_id, syllabus_id, role, and permission context.

Layer	Responsibility
Next.js frontend	Dashboards, forms, file upload UI, AI chat UI, generated paper preview, admin setup flows.
Node.js backend	REST/GraphQL APIs, auth, validation, RBAC, DB operations, signed file URLs, AI orchestration calls.
PostgreSQL	Primary relational data and optionally vector embeddings through pgvector.
Object storage	Stores original PDFs, images, generated paper files, and exports.
AI service	RAG chat, question generation, summarization, OCR-assisted indexing, answer key generation.
Background worker	Async ingestion, AI processing, notifications, retries, cleanup.

7. Data Model Overview

Entity	Important fields
institutions	id, name, type, board_type, address, subscription_plan, status
users	id, institution_id, name, email_or_phone, password_hash, role, status
teachers	user_id, employee_code, department, subjects
students	user_id, admission_no, class_id, section_id
academic_years	id, institution_id, name, start_date, end_date, active
classes	id, institution_id, name, board_type, grade_level
sections	id, class_id, name
subjects	id, institution_id, name, code
syllabus_documents	id, institution_id, board_type, class_id, subject_id, file_url, status, version
sample_question_papers	id, institution_id, class_id, subject_id, syllabus_id, file_url, exam_type, year, status
classrooms	id, institution_id, class_id, section_id, subject_id, teacher_id, syllabus_id, invite_code, status
classroom_members	id, classroom_id, user_id, role, joined_at, status
notes	id, classroom_id, teacher_id, title, description, file_url, file_type, tags, status
bookmarks	id, user_id, note_id
question_paper_jobs	id, classroom_id, teacher_id, input_config_json, status, output_file_url
ai_chatbots	id, classroom_id, syllabus_id, vector_collection_id, status, enabled_for_students

ai_chat_sessions	id, chatbot_id, teacher_id, title, created_at
ai_chat_messages	id, session_id, role, content, citations_json, safety_status
notifications	id, institution_id, user_id, type, title, body, read_at
audit_logs	id, institution_id, actor_id, action, entity_type, entity_id, metadata_json

8. Admin Module Requirements

8.1 Institution Onboarding

Field	Specification
Phase	MVP
Primary users	Institution Admin
Purpose	Allow institution admins to set up Vaasenk for their school/college/coaching center.

Functional requirements

- Admin can enter institution name, type, board type, address, contact person, and academic year.
- Admin can define classes, sections, subjects, and departments.
- Admin can invite teachers and optionally bulk import students using CSV.
- Admin can configure whether teachers are allowed to create classrooms directly.
- Admin can configure whether teachers can upload additional sample papers.

Acceptance criteria

- Institution profile cannot be activated without at least one academic year.
- Admin can create at least one class, section, subject, and teacher.
- All created data is scoped to the institution.
- Duplicate class/section/subject names are prevented within the same institution context.

Edge cases and error states

- If CSV import fails, show row-level validation errors.
- If academic year is inactive, classroom creation should be blocked.
- If subscription is inactive, admin can view data but cannot create new classrooms or upload large files.

UI notes

- Use a step-by-step setup wizard with progress indicator: Institution > Academic Year > Classes > Subjects > Teachers > Syllabus.

8.2 Syllabus Repository and Mapping

Field	Specification
Phase	MVP
Primary users	Institution Admin
Purpose	Reduce teacher burden by letting admins upload common syllabus PDFs once and reuse them across classrooms.

Functional requirements

- Admin can upload syllabus PDFs for Samacheer Kalvi, CBSE, or institution-specific syllabus.
- Admin must tag each syllabus by board, class/grade, subject, academic year, language/medium, and version.
- Admin can mark one syllabus version as active for a class-subject pair.
- Admin can map syllabus to one or many classrooms.
- System must run async processing: parse PDF, OCR if needed, chunk text, generate embeddings, and mark AI-ready status.
- Admin can replace syllabus with a newer version without deleting old generated papers.

Acceptance criteria

- Uploaded syllabus appears in Syllabus Library with status: Uploaded, Processing, AI Ready, Failed, Archived.
- A classroom cannot enable AI chatbot until its mapped syllabus is AI Ready.
- Admin can see which classrooms use each syllabus.
- Replacing syllabus creates a new version and does not silently overwrite old references.

Edge cases and error states

- If PDF is scanned and OCR fails, show Failed OCR status and allow re-upload.
- If syllabus is uploaded without class/subject metadata, block save.
- If syllabus version is archived, existing classrooms can still view old notes but new AI generation must use active version unless manually selected.
- If two admins upload same syllabus, system should warn about possible duplicate.

UI notes

- Use metadata-first upload form. Show file preview, processing timeline, and AI readiness badge.

8.3 Sample Question Paper Repository

Field	Specification
Phase	MVP
Primary users	Institution Admin; optionally Teacher with permission
Purpose	Allow admins to upload previous/sample papers so AI-generated papers follow real exam patterns and do not miss important questions.

Functional requirements

- Admin can upload sample question papers as PDFs/images/documents.
- Each sample paper must be tagged by board, class, subject, exam type, year, term, and linked syllabus if known.
- System parses and indexes sample papers separately from syllabus but links them during question paper generation.
- AI should extract common question patterns, mark distribution, repeated topics, and important concepts.
- Admin can mark sample papers as High Priority, Normal, or Archive.
- Teachers can select whether to use sample paper guidance while generating papers.

Acceptance criteria

- Sample paper upload supports at least PDF in MVP.
- Sample paper shows processing status.
- Generated question paper must display a note: "Generated using syllabus + selected sample paper guidance."
- The system must not copy full sample paper questions blindly unless teacher explicitly asks for practice paper style.

Edge cases and error states

- If sample paper belongs to a different subject/class than selected classroom, do not include it.
- If sample paper quality is poor, show low-confidence extraction warning.
- If no sample papers exist, question generation still works from syllabus only.

UI notes

- Show sample papers as cards with board, subject, class, year, exam type, and AI status.

9. Classroom Module Requirements

9.1 Classroom Creation

Field	Specification
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Phase	MVP
Primary users	Institution Admin; Teacher if allowed
Purpose	Create structured spaces where teachers share notes and students access materials.

Functional requirements

- User can create classroom with class, section, subject, teacher, academic year, and mapped syllabus.
- System generates unique classroom invite code.
- Admin can bulk create classrooms from class-section-subject mapping.
- Teacher can view assigned classrooms after login.
- Student can join classroom using invite code or admin assignment.

Acceptance criteria

- Classroom cannot be created without subject and teacher.
- If syllabus is mapped, the classroom AI chatbot job should be created automatically.
- Classroom code must be unique and revocable.
- Admin can disable/refresh invite code.

Edge cases and error states

- If teacher is removed, classroom must require reassignment.
- If invite code is expired, student receives clear error.
- If syllabus processing fails, classroom remains usable but AI features are disabled until fixed.

UI notes

- Classroom cards should show subject, class, section, teacher, enrolled students, syllabus status, and AI-ready status.

10. Notes Module Requirements

10.1 Teacher Note Upload

Field	Specification
Phase	MVP
Primary users	Teacher
Purpose	Allow teachers to upload classroom materials instantly after writing on the board or preparing files.

Functional requirements

- Teacher can upload image, PDF, or text note to a classroom.
- Teacher can add title, description, tags, and visibility status.
- Teacher can mark a note as Important, Homework, Revision, Formula, Assignment, or Exam.
- System stores original file and creates preview thumbnail where possible.
- Students in the classroom receive notification when note is published.
- Teacher can edit title/description/tags after upload.

Acceptance criteria

- Upload form validates file type and size.
- A note can be saved as Draft or Published.
- Published notes appear immediately in student feed.
- Failed upload does not create broken note cards.
- Teacher can delete own notes unless admin locks content.

Edge cases and error states

- If network fails mid-upload, show retry option.
- If file preview cannot be generated, show generic file icon.
- If classroom is archived, uploading must be blocked.
- If student has not joined classroom, they cannot access note URL directly.

UI notes

- Use a simple upload drawer or full-page form with drag/drop, preview, and Publish button.

10.2 Student Notes Access

Field	Specification
Phase	MVP
Primary users	Student
Purpose	Give students organized access to all classroom notes without using WhatsApp/Discord clutter.

Functional requirements

- Student can view notes from joined classrooms.
- Student can filter by classroom, subject, date, and tag.
- Student can bookmark notes.
- Student can download notes if download is enabled.
- Student can open image/PDF viewer inside web app.
- Student can ask doubt under a note if doubt feature is enabled.

Acceptance criteria

- Student sees only notes from enrolled classrooms.
- Bookmark action updates instantly.
- Downloads are logged for analytics.
- Empty classroom shows friendly message, not blank page.

Edge cases and error states

- If note is deleted after student opens it, show unavailable message.
- If file is too large, use browser preview/download fallback.
- If student is removed from classroom, access must be revoked.

UI notes

- Use clean note cards with thumbnail, title, teacher, tag, date, bookmark, and download action.

11. AI Question Paper Generator Requirements

11.1 Generate Question Paper

Field	Specification
Phase	MVP
Primary users	Teacher
Purpose	Reduce teacher workload by generating exam-ready question papers from mapped syllabus and sample paper guidance.

Functional requirements

- Teacher opens classroom and clicks Generate Question Paper.
- System auto-selects mapped syllabus for that classroom.
- Teacher selects portions: chapter range, topics, units, or manually typed topics.
- Teacher selects exam type: Unit Test, Monthly Test, Quarterly, Half-Yearly, Annual, Revision Test, Custom.

- Teacher configures total marks, duration, difficulty, question types, section pattern, and whether answer key is required.
- Teacher can select sample paper guidance: None, Use all relevant, or choose specific sample papers.
- AI generates a structured paper with title, class, subject, marks, duration, instructions, sections, questions, and optional answer key.
- Teacher can regenerate, edit, save, download PDF, or publish to classroom.

Acceptance criteria

- Generate button disabled until required fields are filled.
- If syllabus is not AI Ready, show “Syllabus is still processing.”
- Generated paper must match selected marks and question types as closely as possible.
- System must show AI confidence or review reminder before teacher downloads/publishes.
- Teacher must be able to edit output before finalizing.
- Generated paper must be saved with input configuration for audit/reuse.

Edge cases and error states

- If AI generation fails, save job as Failed and allow retry.
- If selected portion is not found in syllabus, ask teacher to adjust portion.
- If marks distribution does not add up, block generation or auto-suggest correction.
- If AI produces duplicate questions, system should warn or auto-regenerate duplicates.
- If sample paper conflicts with syllabus, syllabus should be treated as source of truth.

UI notes

- Use a multi-step wizard: Portion > Pattern > Sample Guidance > Generate > Review > Export.
- Show progress states: Preparing syllabus context, analyzing sample papers, drafting paper, formatting output.

11.2 Question Paper Blueprint Inputs

Input	Required?	Example
Classroom	Yes	10-A Mathematics
Syllabus	Auto	Samacheer Kalvi Class 10 Maths 2026
Portion	Yes	Chapter 1 to Chapter 3 or Algebra + Geometry
Exam type	Yes	Unit Test
Total marks	Yes	50 marks
Duration	Optional	1 hour 30 minutes
Question types	Yes	MCQ, 2-mark, 5-mark, 10-mark
Difficulty mix	Optional	30% easy, 50% medium, 20% hard
Sample paper guidance	Optional	Use relevant previous year/sample papers
Answer key	Optional	Generate answer key separately

12. Teacher AI Classroom Chatbot Requirements

This is one of VaasenK's strongest competitive features. Every classroom should have a private AI assistant for the teacher. The assistant should answer using the mapped syllabus and relevant sample question papers. In MVP, students do not get access.

12.1 Classroom AI Assistant Setup

Field	Specification
Phase	MVP
Primary users	Institution Admin, Teacher
Purpose	Automatically create an AI-ready chatbot for each classroom based on the mapped syllabus and sample question papers.

Functional requirements

- When a classroom is created with an AI-ready syllabus, the system creates or links an AI chatbot configuration.
- If the same syllabus and sample paper set is used by many classrooms, the system may reuse the same vector collection but enforce classroom-level permissions.
- Chatbot status can be Not Configured, Processing, Ready, Failed, or Disabled.
- Admin can reprocess syllabus/sample paper data.
- Teacher sees AI assistant only when status is Ready.
- Admin can disable AI assistant for an institution, class, subject, or classroom.

Acceptance criteria

- Teacher cannot access chatbot unless assigned to the classroom.
- Chatbot cannot answer before syllabus indexing is complete.
- Chatbot must keep chat history per teacher and classroom.
- System must log AI usage for cost tracking.
- Chatbot must include disclaimer: "Verify before using in class/exam."

Edge cases and error states

- If no syllabus is mapped, show "Ask admin to map syllabus."
- If sample papers are missing, assistant still works from syllabus.
- If teacher asks outside syllabus, assistant should say it cannot find enough syllabus context.
- If PDF extraction has low confidence, assistant should display limited-confidence warning.

UI notes

- Place AI Assistant as a tab inside Classroom Detail: Notes | Question Papers | AI Assistant | Students | Settings.

12.2 Teacher Chat Capabilities

Field	Specification
Phase	MVP
Primary users	Teacher
Purpose	Help teachers prepare lessons, identify important topics, create quick questions, and answer class doubts using syllabus-grounded AI.

Functional requirements

- Teacher can ask questions about the syllabus: "What topics are in Chapter 4?"
- Teacher can ask for lesson plans: "Prepare a 40-minute teaching plan for Photosynthesis."
- Teacher can ask for important questions based on syllabus and sample papers.
- Teacher can ask for simplified explanations to teach students.
- Teacher can ask for quick quiz questions, homework questions, or revision points.
- Teacher can ask for chapter summary and expected marks weightage if sample paper data is available.
- Chatbot should return answers with source references such as chapter/topic/page if available.
- Teacher can copy response, save response as note draft, or convert response into question paper seed.

Acceptance criteria

- Chat input supports text in MVP.
- Responses should be grounded in syllabus/sample papers.
- Assistant must not fabricate exact page references if unavailable.
- Teacher can start a new chat session or continue previous session.
- Teacher can report bad answer.

Edge cases and error states

- If AI service times out, show retry and keep teacher message unsent or pending.
- If teacher requests student personal data, assistant must refuse unless data module explicitly supports it.

- If generated answer appears beyond syllabus, label it as general knowledge and ask teacher to verify.
- If same question is asked repeatedly, cache may be used but permission checks must still run.

UI notes

- Chat UI should feel like a professional assistant, not a public chat room. Use quick prompts: Important Questions, Lesson Plan, Summary, Quiz, Explain Simply.

12.3 AI Orchestration Recommendation: Flowise vs Langflow

Option	Strength for Vaasenk	Concern	Recommendation
Flowise	Strong fit for visual RAG chatflows, document stores, file uploads, prediction API, and Postgres/pgvector vector storage.	Production behavior must be tested for multi-tenant isolation, streaming reliability, and API ergonomics.	Good for fastest MVP AI assistant and question generation prototype. Keep behind backend.
Langflow	Strong visual builder for RAG flows, API-triggered flows, custom components, and pgvector integration.	May require more Python/service management depending on deployment approach.	Good if the team wants deeper customization and already has Langflow experience.
Custom AI microservice	Maximum control over prompts, schemas, retries, costs, and tenancy.	Slower to build initially.	Recommended long-term if Vaasenk scales heavily.

Locked MVP recommendation: Use the backend-owned AI Orchestration Layer abstraction. Start with either Flowise or Langflow for speed, but design the Node.js backend so Vaasenk can switch AI engine later without changing frontend screens or core product logic.

12.4 RAG Data Boundary Rules

- Every indexed document must carry institution_id, academic_year_id, class_id, subject_id, syllabus_id, and optional classroom_id metadata.
- A teacher query must filter retrieval by teacher permissions and classroom context.
- No classroom should retrieve another institution's syllabus or papers.
- If a shared syllabus collection is reused across classrooms, access must still be checked before query execution.
- Generated answers should cite retrieved chunks where possible.
- The assistant should answer "I do not have enough syllabus context" instead of hallucinating.
- AI prompts must instruct the model to prioritize syllabus over sample papers when conflicts exist.
- Sample papers guide pattern and importance, not blindly copied output.

13. Notifications Requirements

13.1 Web Notifications

Field	Specification
Phase	MVP
Primary users	Admin, Teacher, Student
Purpose	Keep users informed about new classroom activity and AI processing states.

Functional requirements

- System sends notifications for new notes, published question papers, classroom joins, syllabus processing completion/failure, AI paper completion/failure, and teacher doubt replies.
- Users can mark notifications as read.
- Notifications are filtered by role and classroom membership.
- Admin sees system-level processing alerts.

Acceptance criteria

- Notification bell shows unread count.
- Clicking notification navigates to relevant screen.
- Removed users do not receive new classroom notifications.

Edge cases and error states

- If target entity is deleted, notification opens a safe fallback screen.
- If many notifications are generated, batch similar notifications.

UI notes

- Use a top-right notification bell with grouped notification list.

14. Subscription and Pricing Requirements

Plan	Suggested price	Limits	Best for
Starter	₹2,999/month or ₹29,999/year	Up to 500 users, unlimited classrooms, limited AI credits, basic storage	Small schools/coaching centers
Growth	₹5,999/month or ₹59,999/year	Up to 1,500 users, higher AI credits, sample paper repository, basic analytics	Medium schools/colleges
Institution	₹9,999/month+ or custom yearly	Large user base, high storage, high AI usage, onboarding support	Large colleges/institution groups

- Pricing should not be per classroom. One institution payment should allow unlimited classrooms within plan limits.
- AI question generation and chatbot usage should have monthly fair-use credits to control cost.
- Admin dashboard should display AI credit usage clearly.
- For MVP, payment collection can be manual invoice/UPI/bank transfer while product-market fit is tested.

15. Web UI/UX Requirements

Area	Requirement
Design tone	Professional, clean, teacher-friendly, institution-trustworthy, not childish.
Primary palette	Primary Blue #2563EB, Soft Blue #EFF6FF, Navy #0F172A, White #FFFFFF, Light Gray #F8FAFC.
Accent colors	Green #16A34A for success, Amber #F59E0B for important/pending, Red #DC2626 only for danger.
Typography	Inter as primary font; Poppins may be used for logo/marketing headings.
Navigation	Sidebar for web dashboards; top bar for search, notifications, profile.
Components	Cards, tables, tabs, upload dropzones, status badges, modals, wizard forms, file preview, chat panel.
Accessibility	Readable contrast, keyboard navigable forms, visible focus states, alt text for previews.
Empty states	Every empty page must explain what to do next. Example: "No syllabus uploaded yet. Upload syllabus to enable AI features."

16. Web Navigation Map

- Public: Landing Page > Login > Forgot Password.
- Admin: Dashboard > Institution Setup > Classes/Sections > Teachers > Students > Syllabus Library > Sample Papers > Classrooms > Billing > Settings.

- Teacher: Dashboard > My Classrooms > Classroom Detail > Notes > Upload Note > Generate Paper > AI Assistant > Student Doubts > Profile.
- Student: Dashboard > My Classrooms > Notes Feed > Note Detail > Bookmarks > Profile.

17. API Overview

API group	Example endpoints
Auth	POST /auth/login, POST /auth/logout, POST /auth/refresh, POST /auth/invite/accept
Institutions	GET/PUT /institutions/:id, GET /institutions/:id/usage
Users	POST /teachers, POST /students/import, GET /users?role=teacher
Syllabus	POST /syllabus, GET /syllabus, POST /syllabus/:id/reprocess, PATCH /syllabus/:id/archive
Sample papers	POST /sample-papers, GET /sample-papers, POST /sample-papers/:id/reprocess
Classrooms	POST /classrooms, GET /classrooms, POST /classrooms/:id/join, POST /classrooms/:id/refresh-code
Notes	POST /notes, GET /classrooms/:id/notes, PATCH /notes/:id, DELETE /notes/:id
Question papers	POST /question-paper/jobs, GET /question-paper/jobs/:id, POST /question-paper/jobs/:id/publish
AI chatbot	POST /ai/classrooms/:id/chat, GET /ai/classrooms/:id/sessions, POST /ai/classrooms/:id/reindex
Notifications	GET /notifications, PATCH /notifications/:id/read, PATCH /notifications/read-all

18. Security, Privacy, and Compliance Requirements

- Use RBAC for every API endpoint.
- Every database query must be institution-scoped except Super Admin operations.
- Use signed URLs for private files; do not expose raw storage paths.
- Never expose Flowise/Langflow/OpenAI/AI provider API keys to the frontend.
- Store passwords as strong salted hashes.
- Audit critical actions: syllabus upload, paper generation, note deletion, user import, role change.
- Rate-limit login, AI chat, and question generation APIs.
- Sanitize uploaded file names and validate MIME types.
- Scan files if storage provider supports malware scanning or integrate later for institution trust.
- AI logs should avoid unnecessary personal data.

19. MVP Success Metrics

Metric	Target signal
Teacher activation	At least 70% of onboarded teachers upload one note in first week.
Student engagement	At least 60% of students open notes within 24 hours of upload.
AI paper usefulness	At least 50% of teachers generate one question paper in first month.
AI assistant usefulness	At least 30% of teachers use classroom AI assistant weekly after syllabus setup.
Admin onboarding completion	Institution admin completes syllabus + classroom setup without developer help.
Retention	School continues usage after trial/month one.

20. Development Phases

Phase	Features
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Phase 1: UI MVP	Auth screens, dashboards, classroom pages, syllabus library, sample paper library, note upload, note feed, AI question paper UI, AI chatbot UI, settings.
Phase 2: Backend MVP	Auth, RBAC, Postgres schema, file storage, classroom APIs, note APIs, syllabus/sample upload APIs, notifications.
Phase 3: AI MVP	PDF parsing, embeddings, classroom RAG chatbot, question paper generator, worker jobs, AI logs.
Phase 4: Pilot	Onboard one school/coaching center, gather teacher feedback, fix UX and AI quality issues.
Phase 5: Scale	Analytics, payments, OCR improvements, parent login, student AI assistant, mobile offline sync.

21. Final MVP Lock

- Syllabus upload belongs to Admin side during setup and classroom mapping.
- Sample question papers belong to Admin side but teachers may optionally contribute if allowed.
- Teachers should not manually upload syllabus every time they generate a question paper.
- Every classroom should have an AI assistant generated from mapped syllabus/sample papers.
- AI assistant is teacher-only in MVP.
- Students can access notes, downloads, bookmarks, and notifications in MVP.
- Question paper generator is teacher-only in MVP.
- The backend must abstract AI engine choice so Flowise/Langflow can be swapped later.